

Retail Analytics: Predicting Customer Purchase Behavior

Business Introduction:

Welcome to RetailVC's Retail Analytics Solutions, a leading provider of data-driven solutions for retail businesses. RetailVC specializes in using advanced analytics techniques to help retailers optimize their operations, enhance customer experiences, and drive business growth. Its mission is to empower retail organizations with actionable insights that enable them to make informed decisions, improve performance, and remain competitive in today's dynamic marketplace.

Problem Statement:

The goal of this capstone project is to develop a comprehensive understanding of customer behavior and preferences based on retail transaction data. The aim is to analyze customers' demographic information, purchasing patterns, and interactions with marketing campaigns to identify key insights. These insights will inform strategic decision-making and drive business growth. Specifically, the focus will be on predicting customers' responses to marketing campaigns and optimizing promotional strategies to maximize their effectiveness and return on investment (ROI).

Note: For this project, you will use the datasets: `cc_data.csv` and `location_data.csv`. The link to the dataset is provided at the end of the document.

Excel Tasks:

1. Data Exploration:

- Create a **statistical summary** for numerical features
- Create a **line chart** for the number of enrolments by year
- Give a cross-tabulated count for response values against education
- Make a **boxplot** on income and write your observations
- Calculate the age of customers and make a histogram of that

- Visualize the response against Marital_Status

SQL Tasks:

2. Data Loading:

- Create a schema named "**retail_data**"
- Set "**retail_data**" as the default schema
- Create tables to store the retail transaction data
- Set **ct_customer** as the datetime field while loading the data and apply the appropriate date format

3. Data Preprocessing:

- Calculate the total number of customer encounters in the marketing campaign dataset
- Identify the **top 10 most** purchased products in the dataset, such as Wines, Meat Products, etc.
- Find the count of response values
- Determine the **distribution of customers** based on their education level and marital status
- Identify the **average income** of customers who participated in the marketing campaign
- Calculate the total number of promotions accepted by customers in each campaign
- Identify the distribution of customers' responses to the last campaign
- Calculate the average number of children and teenagers in customers' households
- Create an Age column by subtracting **year_birth** from the current year
- Create **Age_group** columns based on the below condition:

WHEN Age BETWEEN 18 AND 25 THEN '18-25'

WHEN Age BETWEEN 26 AND 35 THEN '26-35'

WHEN Age BETWEEN 36 AND 45 THEN '36-45'

WHEN Age BETWEEN 46 AND 55 THEN '46-55'

ELSE '56+'

- Determine the average number of visits per month for customers in each age group

Python Tasks:

4. Exploratory Data Analysis

- **General Data Overview:**
 - Check the first few rows of the dataset to understand its structure
 - Check the data types of each column
 - Check for any missing values in the dataset
- **Descriptive Statistics:**
 - Compute summary statistics for numerical columns (mean, median, min, max, and standard deviation)
 - Explore the distribution of numerical variables using histograms or box plots
- **Univariate Analysis:**
 - Explore the distribution of each numerical variable using histograms or kernel density plots
 - Explore the distribution of each categorical variable using bar plots or pie charts
 - Identify outliers in numerical variables using box plots or scatter plots
- **Bivariate Analysis:**
 - Explore the relationship between numerical variables and the target variable (**Response**) using scatter plots or correlation matrices
 - Explore the relationship between categorical variables and the target variable using bar plots or chi-square tests
 - Explore the relationship between numerical and categorical variables using box plots or violin plots

Observations: Write an analysis report on performing exploratory data analysis (EDA) using Python in the context of building a fraud detection system for Retail Analytics

Tableau Tasks:

5. Visualize Customer Income Distribution grouped by Year of Registration

- Examine the breakdown of education levels and marital status using side-by-side circles to represent count
- Explore the relationship between Income and Wine Spending
- Analyze the frequency of purchases across various product categories
- Create a dashboard using all the visualizations

Input dataset link: [Dataset](#)

[Links to an external site.](#)

Submission

Submit your solution file (.ipynb file format) to get feedback.

Rubric

Capstone (1)

Criteria	Ratings	Pts
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This criterion is linked to a Learning Outcome Project 1: Financial Fraud Detection: Anomaly Detection in Credit Card Transactions Data Preparation and Exploration: Use Excel to create statistical summaries, visualize data, and analyze fraud patterns by gender and category. Load data into a SQL schema, perform initial data exploration, and identify key statistics and trends within the datasets.	5 ptsFull Marks 0 ptsNo Marks	5 pts
This criterion is linked to a Learning Outcome Project 1: Financial Fraud Detection: Anomaly Detection in Credit Card Transactions Advanced Data Analysis and Aggregation: Execute SQL queries to aggregate and analyze transaction data, identifying patterns and correlations, and perform comprehensive exploratory data analysis (EDA) using Python.	5 ptsFull Marks 0 ptsNo Marks	5 pts

This criterion is linked to a Learning Outcome Project 1: Financial Fraud Detection: Anomaly Detection in Credit Card Transactions Visualization and Reporting: Design interactive dashboards in Tableau to visualize transaction data, geographical distributions, and fraud trends, and create detailed reports to communicate insights effectively.	5 ptsFull Marks 0 ptsNo Marks	5 pts
This criterion is linked to a Learning Outcome Project 2: Healthcare Management: Patient Readmission Prediction Data Exploration and Preparation: Create statistical summaries and reports on readmissions and race distribution using Excel. Load and explore the healthcare dataset with SQL, analyzing patient encounters and demographics.	5 ptsFull Marks 0 ptsNo Marks	5 pts
This criterion is linked to a Learning Outcome Project 2: Healthcare Management: Patient Readmission Prediction Perform SQL queries to analyze medications, procedures, readmission rates, and age distributions.	5 ptsFull Marks 0 ptsNo Marks	5 pts

Conduct Python EDA to explore feature correlations, distributions, and identify outliers.		
This criterion is linked to a Learning Outcome Project 2: Healthcare Management: Patient Readmission Prediction Design Tableau dashboards to visualize readmission counts, diagnoses, emergency cases, and age distributions. Use bar graphs, stacked bar charts, and bubble charts to present key insights.	5 ptsFull Marks 0 ptsNo Marks	5 pts
This criterion is linked to a Learning Outcome Project 3: Retail Analytics: Predicting Customer Purchase Behavior Excel Data Exploration: Summarize numerical features, create a line chart for enrolments by year, and visualize response against education and Marital_Status. Use a boxplot to analyze income distribution and create a histogram for customer ages.	5 ptsFull Marks 0 ptsNo Marks	5 pts

This criterion is linked to a Learning Outcome Project 3: Retail Analytics: Predicting Customer Purchase Behavior SQL Data Loading and Preprocessing: Create a "retail_data" schema, load retail transaction data, and set ct_customer as the datetime field. Calculate total customer encounters, identify top purchased products, analyze response counts, and determine customer distributions based on education and marital status.	5 ptsFull Marks 0 ptsNo Marks	5 pts
This criterion is linked to a Learning Outcome Project 3: Retail Analytics: Predicting Customer Purchase Behavior Python EDA: Check dataset structure and data types, handle missing values, and compute summary statistics. Explore numerical and categorical variable distributions, identify outliers, and analyze relationships with the target variable (Response).	5 ptsFull Marks 0 ptsNo Marks	5 pts

Total Points: 45

